E: Project Management

1. Member's Responsibilities

Members	Roles	Description
SANTOS, Freyja	Project Leader	Track tasks, progress, and deadlines, and organize reports and submissions.
RAMOS, Cogie	GUI Programmer	Designs interface, input fields, and run, step, and reset buttons.
LEDESMA, Marc Anton	Simulation Programmer	Develops the population simulation, it includes births, deaths, and resource computation.
BARINAN, Mickaela	Graphs & Reports	Makes the population and resource graphs, generation reports, and the final simulation summary.
BARBA, Raymond	Tester and Documentation	Tests the program, writes the README, and takes documents or screenshots.

2. Task Assignment Matrix

Task	Assigned Members	Expected Output
Project planning and task monitoring	**SANTOS**	Project plan and task progress report
User interface development	**RAMOS**	Interface with run, step, and reset buttons
Simulation rules and calculations	**LEDESMA**	Working population simulation
Graphs and simulation reports	**BARINAN**	Population, resource graphs, and generation reports
Input validation and testing	**RAMOS & BARBA**	Tested and fixed program
System Integration	**RAMOS, LEDESMA, & BARINAN**	Complete and working application

Documentation and screenshots	**SANTOS & BARBA**	README file, algorithm explanation, and screenshots of testing
Final presentation and submission	**[All Members]**	Finished project and demonstration

3. Schedule

Week	**What we will do**	**Submission**
Week 1	Project planning, divide tasks, GUI sketch, decide the simulation rules, and set up GitHub	Progress Report 1
Week 2	Develop the simulation algorithm and test it with sample cases	Progress Report 2
Week 3	Add graphs, build interface, simulation controls, and then combine everything to do system testing	Progress Report 3
Week 4	Fix bugs, finish the documentation, and prepare demo	Final Submission

4. Possible Technical Challenges and Solutions

Problems	**Solutions**
Negative population and resources	Apply limits to prevent values below zero
Incorrect calculation	Write formulas first, and then test them with sample numbers
Problems with run, step, and reset button	Test each button to ensure that it performs the correct action
Invalid inputs	Check every input and show an error message
Incorrect graph	Update the graph using the latest simulation results after each generation